Autonomous Driving Mobility Assignment

16-665: Robot Mobility on Air, Land, and Sea

September 2026

Assignment Parameters

This is an individual assignment. Conceptual collaboration is encouraged but students may not exchange work or code nor copy information from another source. Cite all references. Solutions must be typed using the provided LaTeX submission templates. Submission will be in .pdf form. Points will be given for correct work and correct solutions.

Core and Bonus Point Distribution

Each assignment has core points and bonus points. The current assignment has 10.5 core and 5.5 bonus points. For each assignment, you must submit solutions to the core problems on the assignment due date. You must choose at least 2 out of the 3 assignments for which to do the bonus points and submit solutions to the 2 sets of bonus problems no later than their respective due dates. For this section, you must submit the bonus part by October 20.

Learning Objectives

1. Use analytic techniques to make informed parametric comparisons.
2. Exercise vehicle control strategies in an example maneuver.

Submission Instructions

- Create a .pdf file with your solutions for all core problems, and a .pdf file for bonus problems (if attempted).
- Ensure that your .pdf contains all work, explanations, images, and codes for all the questions.
- Submit to Canvas using the proper core/bonus assignment links by the date specified in the course schedule.

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1 Kinematic Bicycle Model

In class, we derived the full kinematic bicycle model (KBM) based on Rajamani, Chapter 2. The resultant model and accompanying diagram are given below.

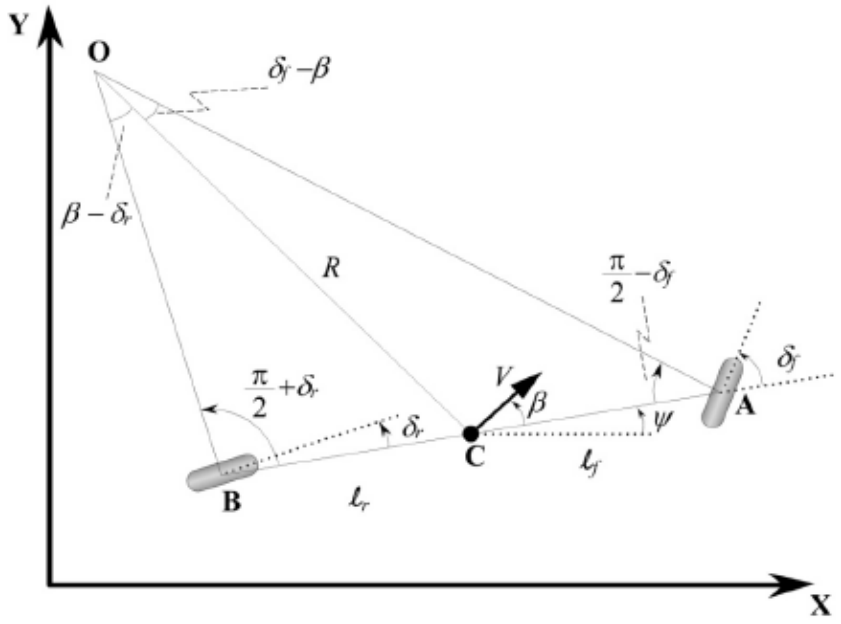


Figure 1: Kinematic Bicycle Model.

$$\dot{X} = V cos(\psi + \beta) \tag{1}$$

$$\dot{Y} = V sin(\psi + \beta) \tag{2}$$

$$\dot{\psi} = \frac{V cos\beta}{l_f + l_r}(tan\delta_f - tan\delta_r) \tag{3}$$

The KBM is usually given in simplified form in research papers. Two examples are given below from Pepy [1] and Kong [2]. The Pepy model measures the position of the vehicle at the rear wheels (Point B in the above diagram), whereas the Kong model uses the center of gravity (Point C in the above diagram). In both cases, a vehicle without steered rear wheels is considered. Derive these models, either by simplifying the Rajamani model, or from first principles. State all assumptions and show all steps.

1.1 Pepy Model [1 point]

The Pepy model:

$$\dot{X} = V cos\psi \tag{4}$$

$$\dot{Y} = V sin\psi \tag{5}$$

$$\dot{\psi} = \frac{V}{l_f + l_r} tan\delta_f \tag{6}$$

1.2 Kong Model [1 point]

The Kong Model:

$$\dot{X} = V cos(\psi + \beta) \tag{7}$$

$$\dot{Y} = V sin(\psi + \beta) \tag{8}$$

$$\dot{\psi} = \frac{V}{l_r} sin\beta \tag{9}$$

1.3 Discussion [2 points]

A. The vehicle slip angle at the center of mass β no longer appears in the Pepy model. Does this mean that the vehicle slip angle is zero? If not, write an equation for it in terms of known/measurable vehicle parameters/variables.

B. What is the difference between vehicle slip and tire slip? Are they related to one another? Can you have one without the other?

1.4 Implementation [3 points]

Implement the Pepy model in Matlab or your favorite software platform with velocity V equal to a constant and zero initial conditions for X, Y, and ψ. Let $l_r = l_f = 1.5$ meters. Use the three different steering inputs δ listed below. Provide one plot and one or two sentences to answer the questions for each part.

A δ = a non-zero constant. Try various constants. Write a mathematical expression for the relationship between the steering angle and the path radius or curvature.

B δ = a sinusoid. Before doing this, try to predict what the trajectory will look like. Did it come out as you expected? Why or why not? Try different velocities V and different amplitudes for the steering sinusoid and comment on the effects.

C δ = a square wave. Why is this unrealistic? How would you handle this lack of realism in the model?

INCLUDE YOUR CODE AT THE END OF YOUR ASSIGNMENT

2 Dynamic Bicycle Model

In class we derived the Dynamic Bicycle Model lateral dynamics based on lateral position error e_1 and yaw angle error e_2:

$$\frac{d}{dt}\begin{bmatrix} e_1 \\ \dot{e}_1 \\ e_2 \\ \dot{e}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & -\frac{2C_{\alpha f}+2C_{\alpha r}}{mV_x} & \frac{2C_{\alpha f}+2C_{\alpha r}}{m} & \frac{-2C_{\alpha f}l_f+2C_{\alpha r}l_r}{mV_x} \\ 0 & 0 & 0 & 1 \\ 0 & -\frac{2C_{\alpha f}l_f-2C_{\alpha r}l_r}{I_zV_x} & \frac{2C_{\alpha f}l_f-2C_{\alpha r}l_r}{I_z} & -\frac{2C_{\alpha f}l_f^2+2C_{\alpha r}l_r^2}{I_zV_x} \end{bmatrix} \begin{bmatrix} e_1 \\ \dot{e}_1 \\ e_2 \\ \dot{e}_2 \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{2C_{\alpha f}}{m} \\ 0 \\ \frac{2C_{\alpha f}l_f}{I_z} \end{bmatrix} \delta + \begin{bmatrix} 0 \\ -\frac{2C_{\alpha f}l_f-2C_{\alpha r}l_r}{mV_x} - V_x \\ 0 \\ -\frac{2C_{\alpha f}l_f^2+2C_{\alpha r}l_r^2}{I_zV_x} \end{bmatrix} \dot{\psi}_{des} \quad (10)$$

2.1 Model Set Up [0 points]

Implement this model in Matlab or your favorite software platform. Use the following typical vehicle parameters: $V_x = 30\ m/sec$, $m = 1573\ kg$, $I_z = 2873\ kg - m^2$, $l_f = 1.1\ m$, $l_r = 1.58\ m$, $C_{\alpha f} = C_{\alpha r} = 80,000\ N/rad$.

The open-loop system, which can be written as $\dot{x} = Ax + B_1\delta + B_2\dot{\psi}_{des}$, has two eigenvalues at the origin and is therefore unstable. You can use state feedback to form the steering input $\delta(t) = -Kx$, leading to $\dot{x} = (A - B_1K)x + B_2\dot{\psi}_{des}$ for the closed-loop system.

2.2 Lane Change [2 points]

For this task, you are asked to switch from following a straight line below the global X-axis to following a straight line along the X-axis.

The vehicle's initial conditions are (X,Y) = (0,-5m) with ψ=0. Vx=30 m/sec as above and remains constant. (In other words, 5m below the origin pointing and traveling in the X-direction.) The vehicle's final conditions are (X,Y) = (100m,0m) with ψ=0. Use a diagonal line to connect the desired Y coordinates over 90 meters in the x axis. The vehicle must stay at Y = -5m until it reaches the diagonal portion of the path at X = 5m. Make sure to add a trajectory along the x-axis at the end so you can examine the maximum error and settling time.

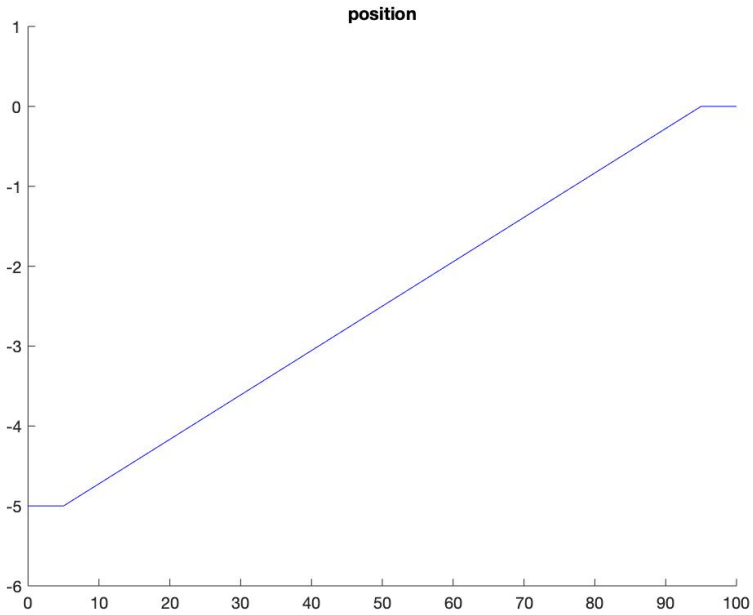


Figure 2: Desired vehicle trajectory

Create a series of inputs $\dot{\psi}_{des}$ that follows this path and use pole placement or LQR to achieve a maximum error abs(e2)≤0.01 rad and reach within abs(e1)≤0.002 m and abs(e2)≤0.0007 rad within 1 second at the transition points (i.e., leaving the initial lane and entering the new lane). State whether you used pole placement or LQR and list the resultant poles. The time step of your simulation should be 0.01 seconds, and the derivative of the steering input $\frac{\delta(t)}{dt}$ should not exceed 25 $\frac{rad}{sec}$. Please include a plot of $\frac{\delta(t)}{dt}$ over the course of the lane change maneuver.

Plot the resultant lateral position error e_1 and yaw angle error e_2 versus time, and make it clear that your simulation meets all of the specifications.

2.3 Desired and Actual Plot [1.5 points]

Plot the true global vehicle path in the XY plane, along with the desired vehicle path. Show separate plots zooming in on the transition points (i.e., leaving the initial lane and entering the new lane).

2.4 Curves [1.5 bonus points]

Create an input $\dot{\psi}_{des}$ corresponding to following a straight path for 1 second, then a positive-curvature circular arc with radius 1000 m for 5 seconds, then a straight path for 1 second, and finally a negative-curvature circular arc with radius 500 for 5 seconds.

Use pole placement or LQR to achieve a maximum error abs(e1)$\leq$0.01 m and abs(e2)$\leq$0.01 rad. State whether you used pole placement or LQR and list the resultant poles.

Plot the resultant lateral position error e_1 and yaw angle error e_2 (these should both converge near 0 when the $\dot{\psi}_{des}$ is 0) versus time.

2.5 Desired and Actual Plot [1 bonus point]

Plot the true global vehicle path in the XY plane, along with the desired vehicle path.

2.6 Discussion [1 bonus point]

Create error plots as in Subsection 2.4 with two different values for V_x. Describe the effect on e_1 and e_2 in one to two sentences.

INCLUDE YOUR CODE AT THE END OF YOUR ASSIGNMENT

3 Pure Pursuit Controller

In Lab 4 of the AD section, you implemented a pure pursuit controller [3] that follows a list of waypoints around the Levine map at a constant speed. In this problem you will measure how fast that controller can complete a lap in simulation, and then augment it so that it laps faster while still tracking the path.

For this problem, please use the provided `lap_timer` ROS 2 package. It times a lap in the F1TENTH simulator and plots the result, and it contains the Levine waypoints that you must follow. Its README explains how to set up and run the lap timer next to your Lab 4 `pure_pursuit` package.

3.1 Augmenting the Pure Pursuit Controller [2 bonus points]

Every lap you report must follow the setup below. The lap timer checks the tracking error and whether the car stopped/crashed and the rest is up to you.

Map and start	Levine, standing start from the default spawn pose $(0, 0, 0)$
Waypoints	`lap_timer/waypoints/levine.csv`, unmodified
Wheelbase	$L_{wb} = 0.3302$ m
Steering angle	$\lvert\delta\rvert \leq 0.4$ rad
Tracking error	at most 0.5 m from the waypoint path during the whole lap

You may change the lookahead distance, the speed, and how either of them is chosen while driving. You will have two versions of the pure pursuit controller:

1. **Vanilla controller:** Use your Lab 4 controller with a constant speed and the steering law $\delta = \arctan(L_{wb}\,\gamma)$ with $\gamma = 2y/L^2$. Here L is the lookahead distance and y is the lateral offset of the goal point in the vehicle frame.

2. **Augmented controller:** Modify your controller so that it completes a valid lap in the Levine map in **10.5 seconds or less**. You will notice that the vanilla controller cannot meet this: at a high constant speed the car leaves the path (or crashes) in the corners, but lowering the speed makes the straights too slow. Based on your understanding of the bicycle model, implement a modifier to the controller that meets both requirements (hint: how are the speed of the vehicle, the curvature of its path and its lateral acceleration related?). If the controller you implemented in Lab 4 already completes a valid lap with this setup under 10.5 seconds, you can use it as

your augmented controller; note that you still need to report a vanilla lap for comparison.

As a reference, our optimized solution for the vanilla controller is able to finish the lap in $\sim$12 s, while the augmented controller is able to lap in $\sim$9 s.

Explain the modifier you implemented and list the equations you used. For both controllers, list the parameters you used and report the lap time and the maximum tracking error printed by the lap timer. Submit the figure produced by `plot_lap` showing the path driven by the vehicle with the waypoints, and the tracking error vs. time.

INCLUDE YOUR CODE AT THE END OF YOUR ASSIGNMENT

Bibliography

[1] *Path Planning Using a Dynamic Vehicle Model*. Proceedings of the 2nd International Conference on Information & Communication Technologies. 2006. URL: https://ieeexplore.ieee.org/abstract/document/1684472/.

[2] *Kinematic and Dynamic Vehicle Models for Autonomous Driving Control Design*. Proceedings of the IEEE Intelligent Vehicles Symposium. 2015. URL: https://ieeexplore.ieee.org/document/7225830/.

[3] *Implementation of the Pure Pursuit Path Tracking Algorithm*. The Robotics Institute, Carnegie Mellon University. 1992. URL: https://www.ri.cmu.edu/pub_files/pub3/coulter_r_craig_1992_1/coulter_r_craig_1992_1.pdf.